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- Under partial correctness, show a formal proof (in Hilbert style) for Conditional Rule 1 given all other rules and axioms in our lecture notes. In other word, given provable triples $\{p \wedge B\} S_1 \{q_1\}$ and $\{p \wedge \neg B\} S_2 \{q_2\}$, prove that $\vdash \{p\} \text{ if } B \text{ then } S_1 \text{ else } S_2 \text{ fi } \{q_1 \vee q_2\}$; and you cannot use Conditional Rule 1 itself in the proof.

Hint: Remember that, in Lecture 11 we have seen that $(B \rightarrow p) \wedge (\neg B \rightarrow q) \Leftrightarrow (B \wedge p) \vee (\neg B \wedge q)$; this logical equivalence can be useful here.

- Let $p \equiv x = 2^k \wedge k \leq n$. Complete the following formal proof by calculating predicates p_1 to p_4 , and completing the rule references R_1 to R_5 . Here, " 2^k " means " 2 to the power of k ".

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|--|-----------------|
| 1. $\{p_1\} x := x * 2 \{p_2\}$ | R_1 |
| 2. $\{p_2\} k := k + 1 \{p_3\}$ | R_2 |
| 3. $\{p_1\} x := x * 2; k := k + 1 \{p_3\}$ | R_3 |
| 4. $p_3 \rightarrow p$ | predicate logic |
| 5. $\{p_1\} x := x * 2; k := k + 1 \{p\}$ | R_4 |
| 6. $\{\text{inv } p\} \text{ while } k < n \text{ do } x := x * 2; k := k + 1 \text{ od } \{p_4\}$ | R_5 |

Hint: Figure out what p_1 is from line 5 and 6 first.

- Complete the following formal proof by completing the rule references R_1 to R_{11} . In the proof, $fac(n)$ is a function and $fac(n) = 1 * 2 * 3 * \dots * n$.

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|--|-----------------|
| 1. $\{n > 0\} k := n - 1 \{n > 0 \wedge k = n - 1\}$ | R_1 |
| 2. $\{n > 0 \wedge k = n - 1\} x := n \{n > 0 \wedge k = n - 1 \wedge x = n\}$ | R_2 |
| 3. $n > 0 \wedge k = n - 1 \wedge x = n \Rightarrow p$
$\# p \equiv 1 \leq k \leq n \wedge x = fac(n) / fac(k)$ | predicate logic |
| 4. $\{n > 0 \wedge k = n - 1\} x := n \{p\}$ | R_3 |
| 5. $\{n > 0\} k := n - 1; x := n \{p\}$ | R_4 |
| 6. $\{p[x * k / x]\} x := x * k \{p\}$ | R_5 |
| 7. $\{p[x * k / x] [k - 1 / k]\} k := k - 1 \{p[x * k / x]\}$ | R_6 |
| 8. $p \wedge k > 1 \Rightarrow p[x * k / x] [k - 1 / k]$ | predicate logic |
| 9. $\{p \wedge k > 1\} k := k - 1 \{p[x * k / x]\}$ | R_7 |
| 10. $\{p \wedge k > 1\} k := k - 1; x := x * k \{p\}$ | R_8 |
| 11. $\{\text{inv } p\} W \{p \wedge k \leq 1\}$
$\# W \equiv \text{while } k > 1 \text{ do } k := k - 1; x := x * k \text{ od}$ | R_9 |
| 12. $\{n > 0\} k := n - 1; x := n \{\text{inv } p\} W \{p \wedge k \leq 1\}$ | R_{10} |
| 13. $p \wedge k \leq 1 \Rightarrow x = fac(n)$ | predicate logic |
| 14. $\{n > 0\} k := n - 1; x := n \{\text{inv } p\} W \{x = fac(n)\}$ | R_{11} |

4. For the formal proof in Question 3:
 - a. Create the corresponding full proof outline under partial correctness.
 - b. Create the corresponding minimal proof outline under partial correctness.
5. Give a full proof outline under partial correctness obtained by expansion of the following minimal proof outline. Please use the **Forward Assignment Axiom** to prove all the assignment statements. Give a brief explanation on each logical implication used in the proof outline.

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{y ≥ 1} x := 0; r := 1;
{inv 1 ≤ r = 2^x ≤ y}
while 2 * r ≤ y do r := 2 * r; x := x + 1 od
{r = 2^x ≤ y ≤ 2^(x + 1)}

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6. Create the formal proof (in Hilbert style) that corresponds to the full proof outline you have created in Question 5.
7. Under total correctness, find a reasonable precondition and create a full proof outline for the following provable triple:

$$\{...\} \text{ if } \text{sqrt}(x) > y \text{ then } x := b[x - y] \text{ else } y := b[y - x] \text{ fi } \{x = y\}$$

Hint: Using Conditional Rule 2 can avoid calculating the domain predicate for a conditional statement.

8. Under total correctness, find a reasonable postcondition and create a full proof outline for the following provable triple:

$$\{\text{sqrt}(x) \leq y\} x := x * y; x := 1 \div x \{...\}$$

Hint: Since the precondition and the statement are not safe, we should include their domain predicates in the precondition for total correctness. Be careful of the domain predicate for a sequence statement.